Area and differential equation (5)

Q1. A solution of the differential equation
$y x^{y-1} dx + x^y \ln x \, dy = 0$ —i

a) $y^x = 1$ (b) $x^y - 1 = 0$ (c) $\frac{1}{y} = \log_2 x$ (d) $x = \log_2 y$.

Q2. Solution If $\phi(x)$ is a differentiable function then the solution of the equation
$dy + [y\phi'(x) - \phi(x)\phi'(x)]dx = 0$ —i

(a) $y = (\phi(x) - 1) + ce^{-\phi(x)}$
(b) $y\phi(x) = (\phi(x))^2 + C$ (c) $ye^{\phi(x)} = \phi(x)e^{\phi(x)} + c$
(d) $(y - \phi(x)) = \phi(x)\, e^{-\phi(x)}$

Q3. If $f(x)$ and $g(x)$ are two solutions of the differential equation $ay'' + x^2y' + y = e^x$ then $f(x) - g(x)$ —i solution of

(a) $a^2y'' + y' + y = e^x$ (b) $ay'' + y = e^x$
(c) $ay'' + x^2y' + y = 0$ (d) $y'' + x^2y' + y = 0$

Q4. The differential equation of the family of hyperbolas with asymptotes as the line $x + y = 1$ & $x - y = 1$ —i

(a) $yy' + x = 0$ (b) $yy' = (x - 1)$
(c) $yy'' + y' = 0$ (d) $y' + xy = 0$

Q5. The solution of $\frac{d^3y}{dx^3} - 8\frac{d^2y}{dx^2} = 0$ satisfying $y(0) = \frac{1}{8}$, $y_1(0) = 0$ & $y_2(0) = 4$ —i

Q6. Solve the differential equation
$$\frac{dy}{dx} = \frac{2(y+2)^2}{(x+y-1)^2}.$$

Q7. Solve: $\frac{dy}{dx} = \frac{y^2 - x}{2y(x+1)}$.

Q8. Show that the differential equation $y^3 dy + (x+y^2)\,dx = 0$ can be reduced to a homogeneous equation. Hence solve it.

Q9. Solve: $3x^2y^2 + \cos(xy) - xy\sin(xy) + \frac{dy}{dx}\{2x^3y - x^2\sin(xy)\} = 0$

Q10. Solve: $\frac{dy}{dx} - y\ln 2 = 2^{\sin x}.(\cos x - 1)\ln 2$, y being bounded when $x \to \infty$.

Q11. Solve $\frac{dy}{dx} + \frac{\cos x(3\cos y - 7\sin x - 3)}{\sin y(3\sin x - 7\cos y + 7)} = 0$

Q12. Obtain the differential equation associated with the primitive $y = c_1 e^{3x} + c_2 e^{2x} + c_3 e^x$.

Q13. Solve: $2y\frac{dy}{dx} = e^{\frac{x^2+y^2}{x}} + \frac{x^2+y^2}{x} - 2x$.

Q14. Let $f: [0,\infty) \to \mathbb{R}$ be a twice differentiable function satisfying $f(0) \geq 0$ & $f'(x) > f(x)$ for all $x > 0$. Prove that $f(x) > 0 \ \forall\, x > 0$.

Q15. Find the curve of the differential equation $x(1 - x\ln y)\frac{dy}{dx} + y = 0$ which passes through $(1, \frac{1}{e})$.

Q16. Find the equation of the curve passing through the origin if the middle point of the segment of its normal from any point of the curve to the x-axis lies on the parabola $2y^2 = x$.

Q17. Let f and g be differentiable functions on the real line satisfying the equation $(f^2+g^2)\cdot f' + (fg)\cdot g' = 0$. Prove that f has maxima.

Q18. An uncommon mistake is to believe that the product rule for derivatives, say that $(fg)' = f'g'$. If $f(x) = e^{x^2}$, determine whether there exists an open interval (a,b) and a non-zero function f defined on (a,b) such that this wrong product rule is true for f and g on (a,b).

Q19. Given two curves $y = f(x)$ passing through the points $(0,1)$ & $y = \int_0^x f(t)\,dt$ passing through the points $(0,0)$. The tangents drawn to both curves at the points with equal abscissas intersect on the x-axis. Find the curve.

Q20. Show that the curve such that the distance between the origin and the tangent at an arbitrary points is equal to the distance between the origin and the normal at the point is $\sqrt{x^2+y^2} = ce^{\pm \tan^{-1}\frac{y}{x}}$.